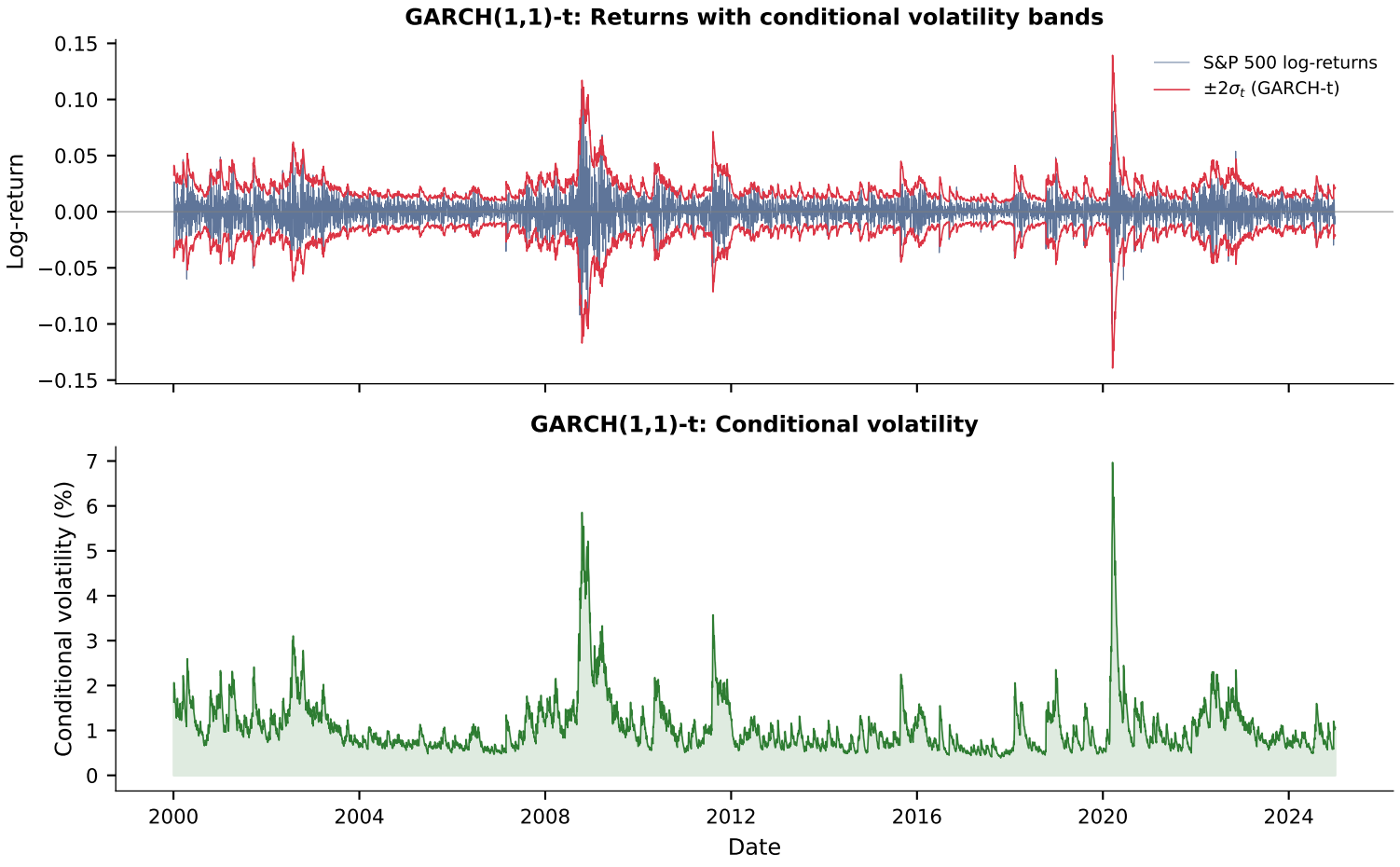
GARCH(1,1)-t: Returns with conditional volatility bands
S&P 500 log-returns
±2σt (GARCH-t)
Log-return
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
GARCH(1,1)-t: Conditional volatility
Conditional volatility (%)
7
6
5
4
3
2
1
0
2000
2004
2008
2012
2016
2020
2024
Date